INTRODUCTION TO DATA MANAGEMENTPROJECT REPORT

(Project Semester January-April 2025)

Coal Stock and Performance Analytics Dashboard for Indian Thermal Power Plants

Submitted by

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Registration No.: 12304191

Programme.: B.Tech. (Computer Science and Engineering)

Section: KM005

Course Code: INT217

Under the Guidance of

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Discipline of CSE/IT

Lovely School of Computer Science and Engineering

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DECLARATION

I, Ashmeet Kumar, student of **B.Tech. (Computer Science and Engineering)** under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 11-04-2025 Signature

Registration No. 12304191 Ashmeet Kumar

<u>CERTIFICATE</u>

This is to certify that Ashmeet Kumar bearing Registration no. 12304191 has completed INT217 project titled, "*District of Columbia Crime Insight.*" under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

Baljinder Kaur

Designation of the Supervisor

School of School of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab.

Date: 11-04-2025

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who supported me throughout the completion of this project, “Coal Stock and Performance Analytics Dashboard for Indian Thermal Power Plants.”

Firstly, I am thankful to my faculty and academic mentor Ms. Baljinder Kaur for her guidance and consistent support. Her insights into data visualization and dashboard design were crucial to the development of this project.

This project allowed me to enhance my skills in Excel-based analytics, particularly in the areas of pivot tables, charting, and interactive dashboard elements. The process helped me convert raw data into useful insights and draw sector-wise comparisons.

Lastly, I would like to thank my friends and family for their encouragement and motivation throughout the development of this project.

TABLE OF CONTENTS

1. INTRODUCTION

This project focuses on building an Excel-based dashboard to analyze coal stock and performance of Indian Thermal Power Plants by sector – Central, State, Private, and Joint Ventures (JV).

Using Microsoft Excel, various metrics like Plant Load Factor (PLF), coal stock, daily requirement, receipt and consumption were analyzed. Through pivot tables, slicers, and charts, sector-level insights were extracted to evaluate efficiency and stock management practices.

The objective is to visually compare and analyze stock handling capabilities and operational efficiency across different ownership sectors.

2. Source of Dataset

The dataset used was provided as a CSV file titled `7773_source_data.csv`, containing coal stock levels, generation figures, PLF, and sector-wise ownership information for thermal power plants across India.

This publicly available dataset to promote transparency and support data-driven decision-making. It can be accessed directly at the following link:
📎 https://ndap.niti.gov.in/dataset/7773

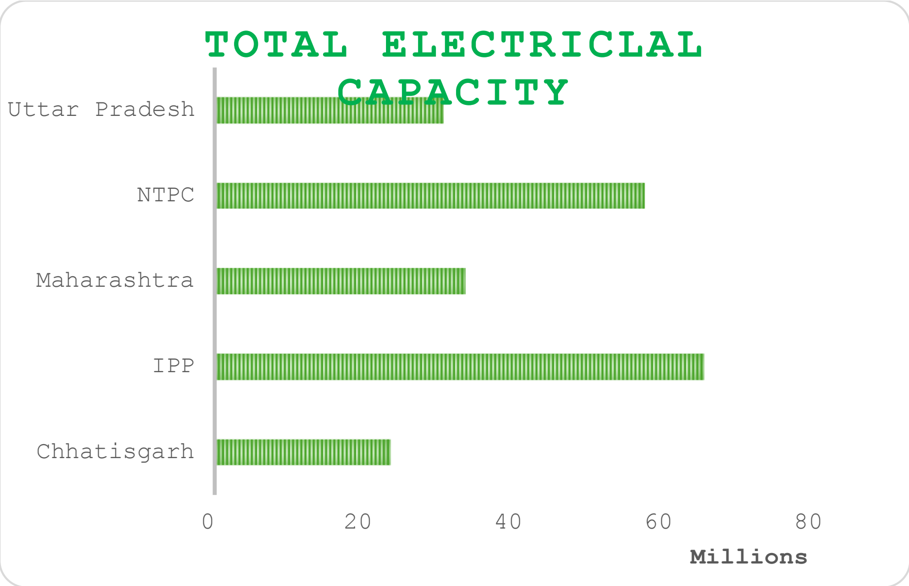
3. Dataset Preprocessing

To prepare the dataset for visualization:
- Renamed key columns for clarity
- Handled missing or inconsistent values
- Ensured consistent format for numeric and date fields
- Verified categorization for sectors: Central, State, Private, JV

4. Analysis on Dataset

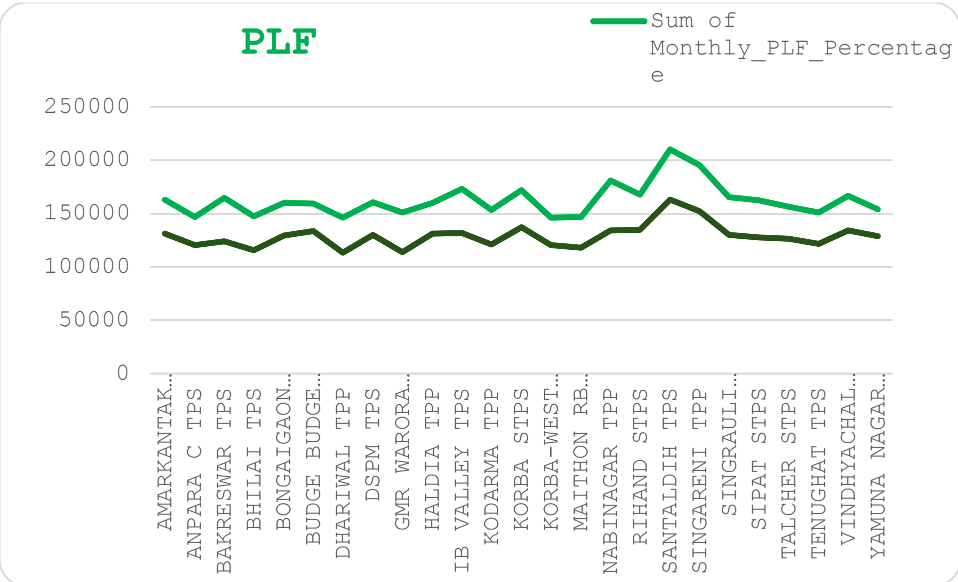
4.1 Electrical Capacity by State

- State players like **Chhattisgarh** and **Maharashtra** contribute significantly but less than Central and IPP entities.



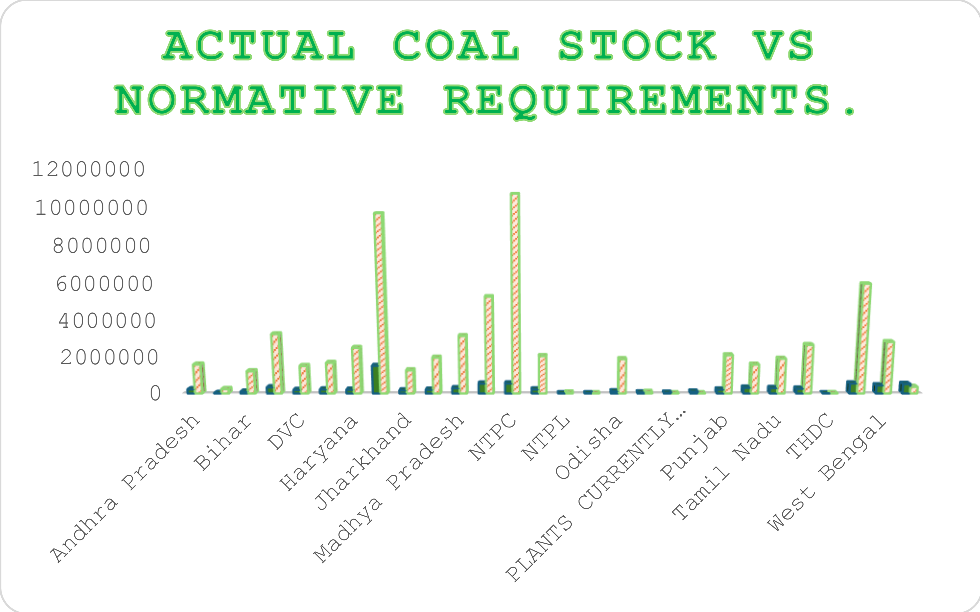
4.2 PLF (Plant Load Factor) Analysis

- Central sector units maintain better load utilization, showing higher operational efficiency.



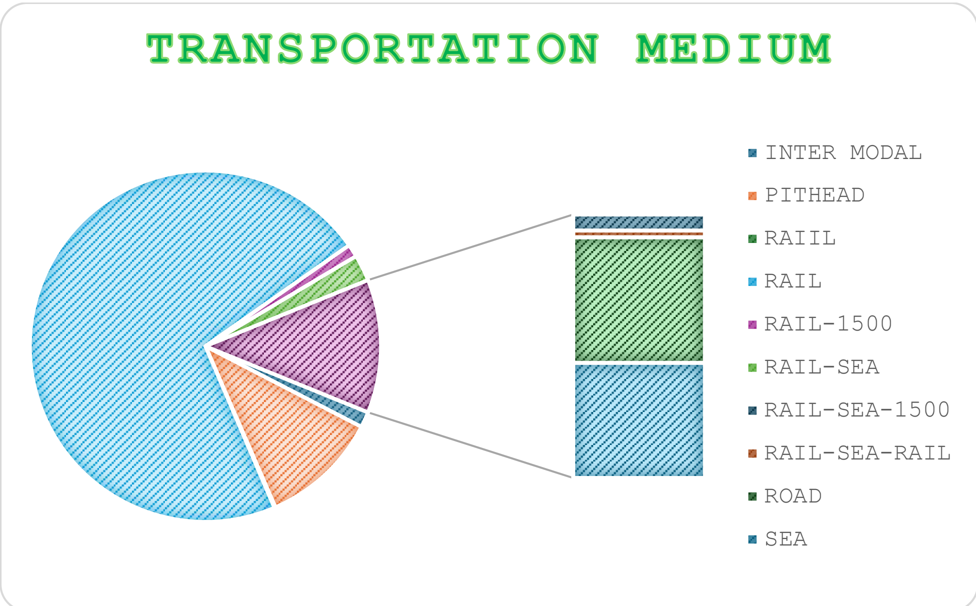
4.3 Stock Requirement vs Actual Stock

- Most Central and larger State plants maintain surplus coal stock. Stock handling is more robust in these sectors.



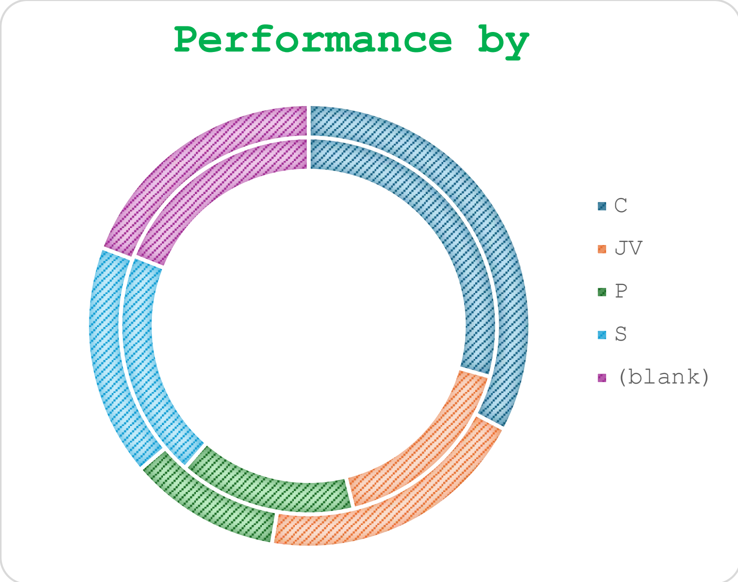
4.4 PLF (Plant Load Factor) Analysis

- Coal logistics heavily rely on rail transport. Cleaning inconsistencies could improve analysis further.



4.5 PLF (Plant Load Factor) Analysis

- Central sector leads in both capacity and stock management. Private sector maintains good efficiency with fewer resources.



5. Conclusion

The Excel dashboard successfully provided sector-wise insights into coal stock and performance for thermal power plants. Central and Private sectors generally showed higher efficiency and better coal handling.

6. Future Scope

- Add time-series data to analyze trends across months/years
- Include regional/geographic filters
- Automate dashboard updates using Power Query or VBA

7. References

- **Open Data DC – Crime Incidents in 2024**
  Government of the District of Columbia. Accessed from:
  https://ndap.niti.gov.in/dataset/7773
- **Microsoft Excel Documentation**
  Microsoft Support: Creating Pivot Charts, Slicers, and Timelines.
  https://support.microsoft.com/excel

8. Social Media Engagement

LinkedIn –
https://www.linkedin.com/posts/ashmeet1346_exceldashboard-dataanalytics-powersector-activity-7317229854502858753-xN0o?utm_source=share&utm_medium=member_desktop&rcm=ACoAAEdCYwQBBlr79GDAuW_6-K-4cG0dZvq5vq4

Dashboard: